ISP CONNECTIVITY WITH Border Gateway Protocol (BGP)

Dual-Homed BGP, NAT/PAT, QoS & Automatic Failover

CISCO Packet Tracer 8.2.2 Platform

1. Project Overview & Business Context

This document details the design, implementation, and live verification of an enterprise network that achieves reliable internet connectivity through two independent ISP providers using Border Gateway Protocol (BGP) for dynamic path selection and automatic failover.

1.1 Business Requirements

An enterprise organization faces three critical and seemingly contradictory requirements: (1) Continuous Internet Access for all employees, (2) Redundancy Against Single Points of Failure, and (3) IP Address Translation from private to public. Solving all three simultaneously requires sophisticated routing, automatic failover, and seamless address translation.

1.2 Why BGP Matters

BGP is the fundamental protocol that holds the entire internet together. Despite its critical importance, most entry-level engineers have never configured it because it requires understanding Autonomous Systems, policy-based routing, and multi-domain network design. Demonstrating working BGP with live failover testing immediately signals real-world expertise far beyond typical entry-level candidates.

2. Border Gateway Protocol Fundamentals

2.1 BGP vs. OSPF

OSPF (used in Project 1) optimizes for routing within a single organization, with all routers trusting each other. BGP optimizes for routing between independent organizations that don't trust each other and may have conflicting goals. BGP finds the administrator-preferred path, not necessarily the mathematically optimal path.

2.2 Autonomous Systems

Every independent network is assigned a unique Autonomous System Number (ASN). Private AS numbers (64512-65534) are for non-internet-facing organizations. In this project: AS 300 (INTERNET-RT), AS 100 (ISP1-RT), AS 200 (ISP2-RT), AS 65001 (EDGE-RT).

3. Network Topology & Devices

3.1 Three-Layer Architecture

Layer 1 (Upstream): INTERNET-RT representing public internet. Layer 2 (ISP): Two independent ISPs (ISP1-RT and ISP2-RT) for redundancy. Layer 3 (Customer): EDGE-RT and internal LANs with users, servers, and phones.

3.2 Device Inventory

Device	Model	Role
INTERNET-RT	2911	Simulated internet
ISP1-RT	2911	Primary ISP (AS 100)
ISP2-RT	2911	Backup ISP (AS 200)
EDGE-RT	2911	Enterprise edge (AS 65001)
LAN-SW1, LAN-SW2	2960	Access switches
ENT-SERVER x2	Server	DHCP/DNS per subnet
PCs, IP Phones	Mixed	Users and voice devices

4. IP Addressing & VLAN Plan

4.1 WAN Links

All WAN links use /30 subnets for point-to-point connections: 203.0.113.0/30 (ISP1↔INTERNET), 203.0.114.0/30 (ISP2↔INTERNET), 100.64.1.0/30 (ISP1↔EDGE), 100.64.2.0/30 (ISP2↔EDGE).

4.2 LAN Subnets

Data VLAN 10 (192.168.10.0/24) and VLAN 20 (192.168.20.0/24) for users. Voice VLANs 110 and 120 (192.168.110-120.0/24) for IP Phones. All private addresses translated to public via NAT/PAT on EDGE-RT.

5. Configuration By Phase

Phase 1: Topology & Addressing

Devices placed, modules installed, WAN IPs assigned, all links up/up.

Phase 2: BGP Peering

```
EDGE-RT BGP config:
router bgp 65001
neighbor 100.64.1.1 remote-as 100
neighbor 100.64.2.1 remote-as 200
network 192.168.10.0 mask 255.255.255.0
network 192.168.20.0 mask 255.255.255.0
network 100.64.1.0 mask 255.255.255.252
network 100.64.2.0 mask 255.255.255.252
```

Both neighbors established with 4 prefixes received each.

Phase 3: NAT/PAT

```
access-list 1 permit 192.168.10.0 0.0.0.255
access-list 1 permit 192.168.20.0 0.0.0.255
interface gig0/0
ip nat inside
interface gig0/1
ip nat inside
interface se0/3/0
ip nat outside
interface se0/3/1
ip nat outside
ip nat inside source list 1 interface se0/3/0 overload
ip nat inside source list 1 interface se0/3/1 overload
```

Enables many-to-one address translation using port numbers.

Phase 4: DHCP/DNS

ENT-SERVER and ENT-SERVER(1) configured with DHCP pools and DNS zones for local hostname resolution.

Phase 5: Dual-Homed BGP Failover (Live Tested)

Continuous ping from PC-HR to INTERNET-RT (203.0.113.1). During ping, ISP1 link shut down. BGP converged to ISP2 within 30 seconds. Ping dropped ~15 packets during reconvergence, then resumed successfully via ISP2.

Phase 6: QoS for VoIP

```
access-list 101 permit udp any any eq 5060
access-list 101 permit udp any any range 16384 32767
class-map VOICE-TRAFFIC
match access-group 101
policy-map WAN-QOS
class VOICE-TRAFFIC
priority 256
class class-default
fair-queue
```

Applied to both WAN interfaces. Voice gets strict priority, data gets fair-queue.

6. Verification & Testing Results

BGP Neighbor Status

```
Neighbor V AS MsgRcvd MsgSent Up/Down State/PfxRcd
100.64.1.1 4 100 13 3 00:01:11 4
100.64.2.1 4 200 14 2 00:00:13 4
```

Both neighbors established, exchanging routes.

NAT Translations

```
Pro Inside global Inside local Outside local Outside global
icmp 100.64.1.2:26 192.168.10.10:26 203.0.113.1:26 203.0.113.1:26
icmp 100.64.1.2:27 192.168.10.10:27 203.0.113.1:27 203.0.113.1:27
icmp 100.64.1.2:2 192.168.20.11:2 203.0.113.1:2 203.0.113.1:2
```

Multiple internal IPs translated to public IP 100.64.1.2 via PAT.

QoS Policy

Strict Priority queue at 256 kbps for voice traffic. Fair-queue for data.

Live Failover Test (Critical Proof)

Ping ran continuously. When ISP1 shut down: BGP detected failure, converged to ISP2 in ~30 seconds, ping resumed successfully. show ip bgp confirmed path shifted from AS 100 to AS 200. show ip nat translations showed NAT switched to ISP2 serial IP.

7. Key Insights

Why This Matters

Dual-homed BGP eliminates single points of failure. When configured correctly, your network stays online even if an entire ISP infrastructure fails. This is why enterprises pay for dual connections.

BGP Decision Process

When multiple paths exist, BGP selects based on: (1) Local Preference (highest wins), (2) AS-PATH (shortest wins), (3) Origin, (4) MED, (5) eBGP over iBGP, (6) Router ID. ISP1 naturally preferred due to shorter AS-PATH.

NAT/PAT Bridge

Private internal networks use RFC 1918 addresses. Public internet only understands public IPs. NAT translates between them. PAT allows thousands of internal devices to share one public IP via port-based session tracking.

QoS Critical for Voice

Without QoS, data transfers monopolize WAN bandwidth, causing voice packets to drop and calls to sound robotic. QoS ensures voice always gets priority queuing, keeping calls crisp.

8. Skills Demonstrated

BGP configuration (eBGP peering, neighbors, networks) • Multi-ISP redundancy design • NAT/PAT configuration • WAN link management • QoS architecture • DHCP/DNS • Voice VLAN design • Live failover testing • Routing table analysis • NAT translation verification • Network troubleshooting and documentation.